

Align Filter and Non-Holonomic Mount Observations

sensor_fusion documentation

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1 Scope

This document records the mathematical model implemented by `sensor_fusion/src/align/mod.rs`. The align filter is a standalone IMU-to-vehicle mount estimator. It estimates the mount quaternion and its three-dimensional small-angle covariance. The runtime fusion pipeline uses this mount as the coarse seed for EKF initialization.

2 Frames and Quaternion Convention

Symbol	Meaning
n	Local navigation frame, NED: north, east, down. GNSS velocities supplied to align are in this frame.
b	Raw IMU body frame. Mean gyroscope and accelerometer samples are expressed in this frame.
v	Vehicle frame, FRD: forward, right, down. The non-holonomic assumption constrains lateral and vertical vehicle velocity.

The project math notes use active rotation matrices

$$x_a = C_{ab}x_b, \quad (1)$$

where C_{ab} maps a vector from frame b into frame a . Quaternions use the same direction:

$$R(q_{ab}) = C_{ab}, \quad R(q_1 \otimes q_2) = R(q_1)R(q_2). \quad (2)$$

The align state stores the public physical vehicle-to-body mount q_{bv} . Its rotation matrix is $C_{bv} = R(q_{bv})$:

$$x_b = C_{bv}x_v. \quad (3)$$

The body/sensor-to-vehicle rotation is the transpose:

$$x_v = C_{vb}x_b = C_{bv}^T x_b. \quad (4)$$

For small-angle corrections, the implementation uses

$$\delta q(\delta\theta) \simeq \begin{bmatrix} 1 & \frac{1}{2}\delta\theta^T \end{bmatrix}^T. \quad (5)$$

General roll/pitch/yaw mount corrections are left-injected,

$$q_{bv}^+ = \delta q(\delta\theta) \otimes q_{bv}^-, \quad (6)$$

while the horizontal heading update applies a vehicle-frame yaw correction on the right,

$$q_{bv}^+ = q_{bv}^- \otimes q_z(\delta\psi_v). \quad (7)$$

3 State and Covariance

The nominal state is only the mount quaternion:

$$x = q_{bv} \in \mathbb{S}^3. \quad (8)$$

The covariance is defined on a three-state small-angle error:

$$\delta x = \begin{bmatrix} \delta\phi & \delta\theta & \delta\psi \end{bmatrix}^T \in \mathbb{R}^3, \quad P \in \mathbb{R}^{3 \times 3}. \quad (9)$$

The state order is roll, pitch, yaw of the mount error. Standard-deviation configuration fields are one-sigma values; the filter stores variances in P .

4 Stationary Initialization

Given stationary accelerometer samples $f_{b,k}$, the bootstrap mean is

$$\bar{f}_b = \frac{1}{N} \sum_{k=1}^N f_{b,k}. \quad (10)$$

At rest, the measured specific force is approximately opposite vehicle down, so the vehicle down axis expressed in body coordinates is initialized as

$$\hat{z}_v^b = -\frac{\bar{f}_b}{\|\bar{f}_b\|}. \quad (11)$$

The forward axis is chosen by projecting a reference body x -axis into the plane orthogonal to $\hat{z}_v^b$:

$$\tilde{x}_v^b = x_{\text{ref}} - \hat{z}_v^b (\hat{z}_v^b)^T x_{\text{ref}}, \quad \hat{x}_v^b = \frac{\tilde{x}_v^b}{\|\tilde{x}_v^b\|}. \quad (12)$$

If the projection degenerates, the code switches to a body y -axis reference. The lateral axis is

$$\hat{y}_v^b = \frac{\hat{z}_v^b \times \hat{x}_v^b}{\|\hat{z}_v^b \times \hat{x}_v^b\|}, \quad \hat{x}_v^b \leftarrow \frac{\hat{y}_v^b \times \hat{z}_v^b}{\|\hat{y}_v^b \times \hat{z}_v^b\|}. \quad (13)$$

The tilt-only mount matrix uses these axes as body-frame columns:

$$C_{bv}^{\text{tilt}} = \begin{bmatrix} \hat{x}_v^b & \hat{y}_v^b & \hat{z}_v^b \end{bmatrix}. \quad (14)$$

A supplied yaw seed is then applied as a vehicle-frame yaw delta:

$$C_{bv,0} = C_{bv}^{\text{tilt}} C_z(\psi_{\text{seed}} - \psi_{\text{tilt}}). \quad (15)$$

After initialization the covariance is reset to conservative roll, pitch, and yaw uncertainty:

$$\sigma_{\phi,\theta,\psi,0} = \begin{bmatrix} 10^\circ & 10^\circ & 60^\circ \end{bmatrix}. \quad (16)$$

This keeps the stationary gravity seed useful without treating the initial tilt estimate as calibrated mount accuracy.

5 Prediction

Align prediction does not mechanize attitude with gyro integration. The nominal mount is constant:

$$q_{bv,k+1}^- = q_{bv,k}^+ \quad (17)$$

Only the diagonal covariance receives random-walk process noise:

$$P_{ii,k+1}^- = P_{ii,k}^+ + \sigma_{q,i}^2 \Delta t, \quad i \in \{\phi, \theta, \psi\}. \quad (18)$$

6 Shared Observation Linearization

For a window mean gyroscope ω_b and a body-frame acceleration-like vector a_b , the code predicts vehicle-frame channels as

$$h(q_{bv}) = \begin{bmatrix} \omega_v \\ a_v \end{bmatrix} = \begin{bmatrix} C_{vb}\omega_b \\ C_{vb}a_b \end{bmatrix}, \quad C_{vb} = C_{bv}^T. \quad (19)$$

With the implemented left small-angle perturbation of q_{bv} , the generated Jacobian rows used by the Kalman updates are

$$H_\omega = C_{vb} [\omega_b]_\times, \quad H_a = C_{vb} [a_b]_\times. \quad (20)$$

Masked rows zero selected columns when an observation should not update yaw or when yaw receives an extra scalar scale.

Scalar and two-row updates use standard linear Kalman equations:

$$r = z - h(\hat{x}), \quad (21)$$

$$S = HPH^T + R, \quad (22)$$

$$K = PH^T S^{-1}, \quad (23)$$

$$\delta x = Kr, \quad (24)$$

$$P^+ \leftarrow (I - KH)P^-. \quad (25)$$

After each update, the small angle is injected into q_{bv} and P is symmetrized. The implementation uses the simplified covariance update above, not the Joseph form.

7 Stationary Gravity Observation

Stationary windows are detected by low gyro norm, accelerometer norm near standard gravity, and near-zero GNSS speed. During such windows, the low-pass gravity vector is updated:

$$g_{b,lp}^+ = (1 - \alpha)g_{b,lp}^- + \alpha \bar{f}_b. \quad (26)$$

The gravity observation rotates this vector into the vehicle frame:

$$g_v = C_{vb}g_{b,lp}. \quad (27)$$

For a level stationary vehicle, the constrained channels are

$$g_{v,y} \approx 0, \quad g_{v,z} \approx -\|g_{b,lp}\|. \quad (28)$$

The current implementation applies the y and z rows with a roll/pitch mask, so yaw is not directly updated by stationary gravity.

8 GNSS-Derived Horizontal Heading Observation

For consecutive GNSS velocities $v_{n,k-1}$ and $v_{n,k}$,

$$a_n = \frac{v_{n,k} - v_{n,k-1}}{\Delta t}. \quad (29)$$

Using the midpoint horizontal velocity direction

$$\hat{t} = \frac{[v_N, v_E]^T}{\sqrt{v_N^2 + v_E^2}}, \quad \hat{\ell} = \begin{bmatrix} -\hat{t}_E & \hat{t}_N \end{bmatrix}^T, \quad (30)$$

the GNSS longitudinal and lateral accelerations are

$$a_{\text{long}} = \hat{t}^T a_{n,NE}, \quad a_{\text{lat}} = \hat{\ell}^T a_{n,NE}. \quad (31)$$

The body accelerometer vector is first stripped of its gravity-axis component. Using the current mount, the body gravity axis is

$$\hat{g}_b = -C_{bv} e_z, \quad (32)$$

and the horizontal body acceleration is

$$a_{b,h} = \bar{f}_b - \hat{g}_b \hat{g}_b^T \bar{f}_b. \quad (33)$$

It is then expressed in the vehicle frame:

$$a_{v,h} = C_{vb} a_{b,h}. \quad (34)$$

When speed and acceleration gates pass, align compares the two horizontal directions by

$$e_\psi = \text{atan2}(a_{v,h,x} a_{\text{lat}} - a_{v,h,y} a_{\text{long}}, a_{v,h,x} a_{\text{long}} + a_{v,h,y} a_{\text{lat}}). \quad (35)$$

The residual is treated as a vehicle-frame yaw observation:

$$\delta\psi_v = -K_\psi e_\psi, \quad K_\psi = \frac{P_{\psi\psi}}{P_{\psi\psi} + R_\psi}. \quad (36)$$

The yaw correction is right-injected as $q_{bv}^+ = q_{bv}^- \otimes q_z(\delta\psi_v)$. Roll/yaw and pitch/yaw cross-covariances are zeroed after this scalar yaw update.

The yaw observation variance is not only a fixed tuning value. The implementation combines:

- configured model noise for unmodeled road and suspension effects,
- GNSS acceleration covariance propagated from previous/current GNSS velocity one-sigma values,
- IMU mean-acceleration covariance scaled by the number of IMU samples in the window,
- current tilt covariance propagated through the horizontal acceleration projection Jacobian.

This keeps the yaw update as a one-dimensional observable-subspace correction while making the confidence respond to source uncertainty.

9 Turn Consistency and Planar-Gyro Observation

Turn windows require minimum speed, course rate, and lateral acceleration. Before turn-heading observations receive nominal trust, a ring buffer checks

$$a_{\text{lat}} \approx v \dot{\chi} \quad (37)$$

for sign and magnitude consistency over several retained windows.

When the turn gyro path is enabled, the body gyro mean is rotated into vehicle coordinates:

$$\omega_v = C_{vb} \bar{\omega}_b. \quad (38)$$

The planar-motion constraint is

$$\omega_{v,x} \approx 0, \quad \omega_{v,y} \approx 0. \quad (39)$$

By default this update affects roll and pitch. It may update yaw only if the Yaw is intentionally not updated by this path; yaw information is supplied by the horizontal heading observation above.

10 Readiness and Known Limitations

The align filter reports coarse alignment ready after yaw has been observed and the one-sigma roll, pitch, and yaw covariances are below

$$\sigma_{\text{ready}} = \begin{bmatrix} 5^\circ & 5^\circ & 8^\circ \end{bmatrix}. \quad (40)$$

When automatic mount mode initializes the EKF, the default runtime behavior is to copy align's 3×3 local mount covariance into the EKF residual mount state covariance. The visualizer exposes this as the *Use align mount covariance* option so the handoff can be compared against fixed EKF mount sigma settings.

Important limitations of the current model are:

- Gravity observes roll and pitch only during stationary windows.
- Horizontal acceleration is projected into a heading/yaw residual; it does not directly solve a complete vector mount least-squares problem.
- The mount process model is a random walk with no vehicle dynamics.
- The yaw update is a scalar vehicle-frame correction, while other mount updates use left-injected body-frame small angles.